

Data Engineering Fundamentals and API-Driven Pipelines

Topics Covered

1. Introduction to Data Engineering
2. Data Sources and Data Lifecycle
3. Data Ingestion Patterns (Batch, Stream, File-Based)
4. ETL / ELT Process
5. Data Validation and Data Transformation
6. Data Storage, Data Lakes, and Data Warehousing
7. Data Modeling and Data Pipelines
8. Data Quality, Governance, and Analytics Applications
9. Data Engineering Tools and Technologies
10. REST APIs and JSON Fundamentals
11. Tools & Frameworks for API Testing and Pipeline Development
12. Pipeline Design for AI Systems
13. API-to-Model Workflow Lab
14. Pipeline Documentation and Business Impact Assessment

Outcome Mapping

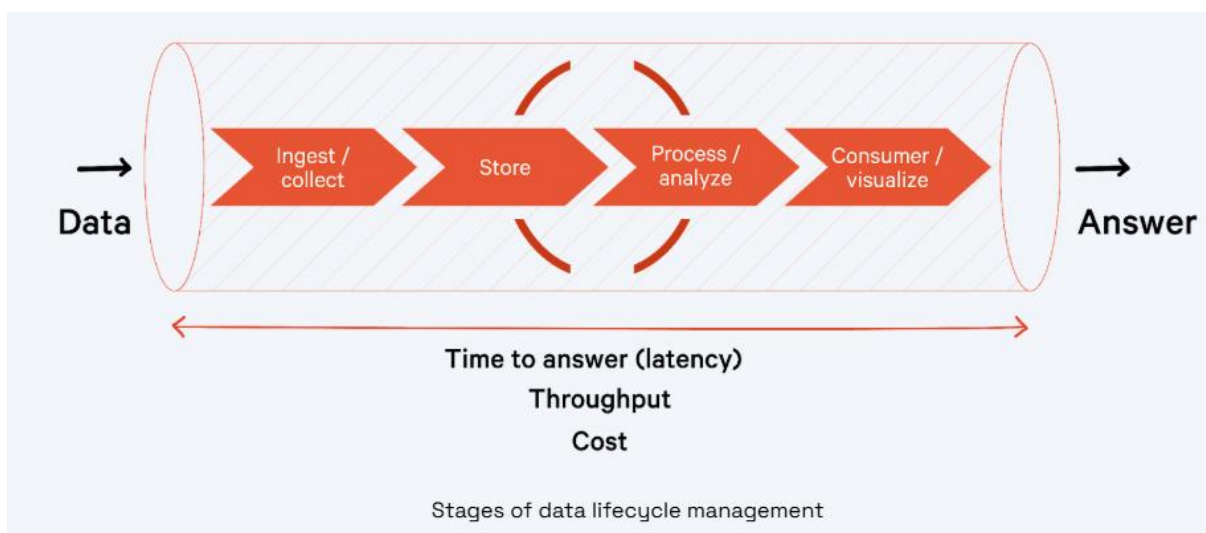
Session Outcome	Topics Covering It
Understand the core concepts of APIs, Data Engineering & Pipelines and relate them to the business use case	Topics 1-8
Use tools and frameworks to implement and test a guided build	Topics 9-10
Translate APIs, Data Engineering & Pipelines into a deployable or reviewable artifact with business relevance	Topics 10-11

Data Life Cycle Management

Data Lifecycle Management (DLM) is the process of managing data throughout its entire journey, from creation and collection to storage, processing, usage, and eventual archival or deletion.

Objectives:

- Ensure data is accurate and reliable.
- Make data easily accessible when needed.
- Protect data from unauthorized access and loss.
- Support efficient data management throughout its lifecycle.



Data Engineering Fundamentals

- Data Engineering is the process of collecting, cleaning, transforming, and organizing data so it can be used for analysis, reporting, and AI applications.
- Data Engineering turns raw data into useful information for business decisions.



Data Engineering Life Cycle



Building Blocks of Data Engineering

1. Data Sources
2. Data Ingestion
3. ETL / ELT Processes
4. Data Storage
5. Data Lakes
6. Data Warehouses
7. Data Modeling
8. Data Pipelines
9. Data Quality & Validation
10. Analytics & AI Applications

Data Sources

The origins from which data is collected for storage, processing, and analysis.

Examples:

- Databases (MySQL, PostgreSQL, SQL Server)
- APIs and Web Services
- CRM & ERP Systems
- CSV, Excel, and JSON Files

- Websites and Social Media
- IoT Devices and Sensors
- Cloud Applications (Salesforce, SAP, etc.)

Purpose:

Provide the raw data required for analytics, reporting, and AI applications.

Data Ingestion

The process of collecting and moving data from different sources into a storage or processing system.

Types:

1.Batch Ingestion

Data is collected and processed at scheduled intervals.

Characteristics

- Processes large volumes of data at once.
- Data size is usually known and finite.
- Results are generated after the entire batch is processed.
- Suitable when real-time insights are not required.

Examples

- Payroll processing
- Monthly billing systems
- Sales reports generated at the end of the day
- Data warehouse updates

Advantages

- Efficient for handling large datasets.
- Lower operational costs.
- Easier error tracking and auditing.
- Can be scheduled during off-peak hours.

Limitations

- Delayed availability of results.

- Less flexible when new data arrives during processing.

2.Stream Processing (Real-Time Processing)

Data is processed continuously as it is generated, allowing organizations to react immediately to new information.

Characteristics

- Processes data in real time or near real time.
- Handles continuously generated data streams.
- Provides immediate results and insights.
- Suitable for time-sensitive applications.

Examples

- Fraud detection in banking
- Stock market monitoring
- IoT sensor data analysis
- Social media analytics

Advantages

- Immediate insights and faster decision-making.
- Supports continuous monitoring of systems and events.
- Highly scalable for large and growing data streams.
- Enables real-time alerts and notifications.

Limitations

- More complex to implement and maintain.
- Requires higher computing and infrastructure resources.
- Can be more expensive than batch processing.

Choosing the Right Ingestion Approach

Use Batch Processing When:

Large amounts of historical data need to be processed, Real-time insights are not critical, Cost efficiency is important.

Use Stream Processing When:

Immediate response is required, Data is generated continuously, Real-time monitoring and analytics are needed.

File-Based Ingestion

- Data is collected from files such as CSV, Excel, JSON, XML, and logs.
- Commonly used when APIs or real-time connections are unavailable.
- Suitable for periodic data transfers and reporting processes.

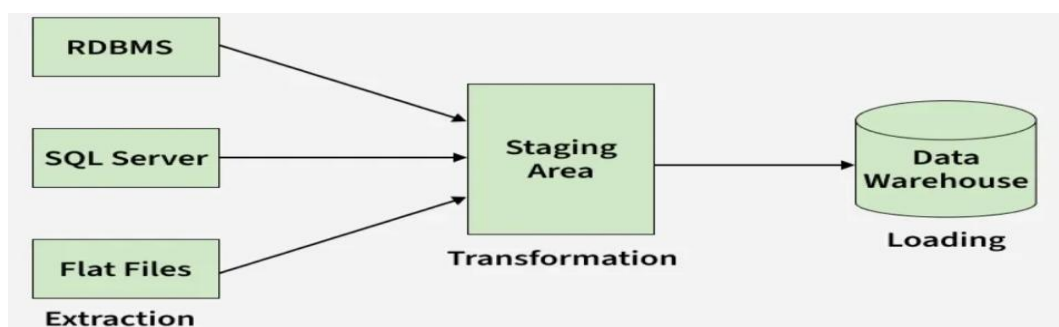
Examples

- Importing sales data from CSV files
- Uploading Excel reports
- Processing JSON/XML files from external systems

ETL Process

ETL is a data integration process that collects data from different sources, refines it into a consistent structure, and stores it in a centralized repository for reporting and analysis.

- **Extract:** Gather data from multiple systems and sources.
- **Transform:** Clean, standardize, and prepare the data for use.
- **Load:** Store the processed data in a data warehouse or data lake.



Extract Phase

The **Extract** step is the starting point of the ETL process, where data is collected from different sources and brought into the pipeline for further processing. At this stage, the data is gathered in its original form without making any changes.

Common Data Sources:

- **Structured Data:** SQL databases, ERP systems, CRM applications
- **Semi-Structured Data:** JSON files, XML files
- **Unstructured Data:** Emails, web pages, text files, documents

Purpose: Gather data from multiple sources and make it available for transformation and analysis.

Transform Phase

The **Transform** step converts raw data into a clean, organized, and usable format. During this stage, data is processed according to business requirements to improve its quality and consistency before analysis.

Common Transformation Activities:

- **Data Filtering:** Remove unnecessary, incorrect, or duplicate records.
- **Data Sorting:** Organize data in a structured order for easier processing.
- **Data Aggregation:** Combine and summarize data to generate useful insights.
- **Data Standardization:** Format data consistently across different sources.
- **Data Validation:** Check and correct data quality issues.

Purpose: Prepare high-quality data that is accurate, consistent, and ready for reporting, analytics, and decision-making.

Load Phase

The **Load** step is the final stage of the ETL process, where the prepared data is stored in a target system such as a data warehouse or data lake. Once loaded, the data becomes available for reporting, analytics, and business decision-making.

Loading Methods:

- **Full Load:** Transfers all available data into the target system, typically used during the initial setup.
- **Incremental Load:** Transfers only new or modified records, making updates faster and more efficient.

Purpose: Store processed data in a centralized location so it can be easily accessed and analyzed.

ETL Pipelining

ETL pipelining improves efficiency by allowing the **Extract, Transform, and Load** stages to run simultaneously instead of waiting for one stage to finish before starting the next.

- Data can be transformed as soon as it is extracted.
- Processed data can be loaded while new data is still being collected.
- Parallel execution reduces processing time and improves performance.
- Better utilization of system resources makes the pipeline more scalable.

Importance of ETL

- **Data Integration:** Combines data from multiple sources into a unified view.
- **Improved Data Quality:** Cleans and standardizes data for accurate analysis.
- **Supports Data Warehousing:** Prepares data for storage and reporting.
- **Better Decision-Making:** Enables insights, forecasting, and strategic planning.
- **Increased Efficiency:** Automates data movement and processing.

Common ETL Challenges

- **Data Quality Issues:** Missing, duplicate, or inconsistent records.
- **Performance Bottlenecks:** Large datasets can slow processing.
- **Scalability Limitations:** Growing data volumes require more robust solutions.

Solutions to ETL Challenges

- **Data Validation & Cleansing:** Improve data accuracy and consistency.
- **Performance Optimization:** Use parallel processing and batch execution.
- **Scalable Platforms:** Leverage cloud-based ETL solutions for growth.

Popular ETL Tools

Tool	Key Feature	Best Used For
Apache NiFi	Real-time data flow and automation	Real-time data ingestion, IoT data, and system-to-system data movement.
Talend	Supports batch and real-time integration	Integrating data from multiple sources and enterprise ETL projects.
Microsoft SSIS	High-performance enterprise ETL	SQL Server-based data integration and large-scale enterprise data warehousing.
Hevo	Automated cloud data pipelines	Moving data to cloud data warehouses with minimal coding and maintenance.
Oracle Warehouse Builder	Data warehouse integration and transformation	Oracle ecosystem projects and enterprise data warehouse implementations.

Choosing the Right ETL Tool

- **Data Volume:** Select scalable tools for large datasets.
- **Real-Time Needs:** Choose tools that support streaming and real-time processing.
- **Budget:** Open-source tools are cost-effective for smaller organizations.
- **Ease of Use:** Commercial tools often provide user-friendly interfaces.

Data Storage

The process of saving and managing data in a centralized location for future access, processing, and analysis.

Types:

- **Databases:** Store structured operational data.
- **Data Lakes:** Store large volumes of raw data in different formats.
- **Data Warehouses:** Store processed and structured data for analytics.

Examples:

- MySQL
- PostgreSQL

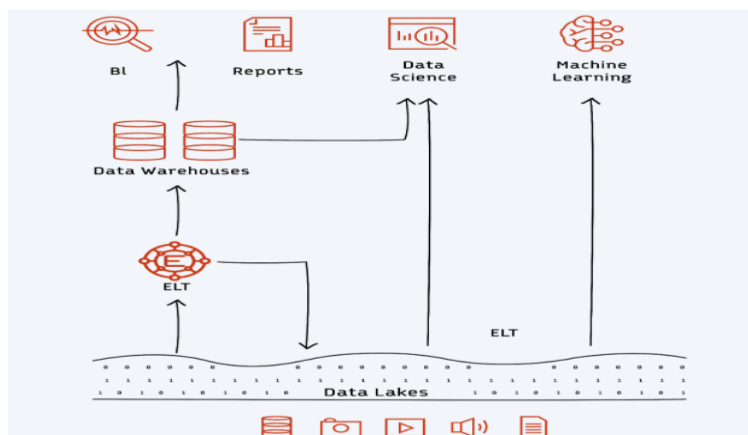
- Amazon S3
- Azure Data Lake Storage
- Snowflake
- Amazon Redshift

Purpose:

Provide a secure, scalable, and reliable location to store data for reporting, analytics, and AI applications.

Data Lakes

A **Data Lake** is a centralized storage system that stores **structured, semi-structured, and unstructured data** in its original (raw) format



Key Features

- Stores all types of data in one place.
- Uses a **store first, analyze later** approach.
- Supports **schema-on-read** (structure is applied during analysis).
- Highly scalable for large volumes of data.
- Cost-effective for storing raw data.

Common Technologies

- Hadoop HDFS
- Amazon S3
- Azure Data Lake Storage

Data Lake Zones

Raw Zone (Landing Zone)

- Stores data exactly as received.
- No transformations or cleaning applied.

Cleansed Zone

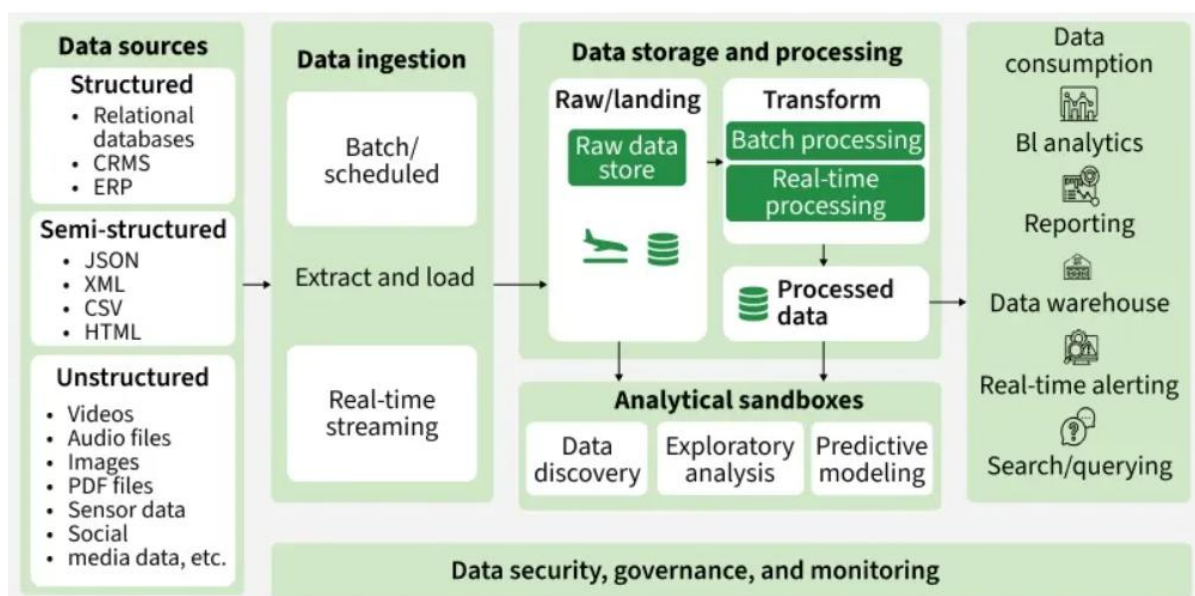
- Data is cleaned, validated, and standardized.
- Improves consistency and quality.

Curated (Trusted) Zone

- **Stores processed and analytics-ready data.**
- Optimized for reporting, dashboards, and AI applications.

Sandbox Zone

- Used by analysts and data scientists for experimentation and testing.



Performance Optimization Techniques

- **Partitioning:** Organize data by date, region, or category for faster access.
- **Columnar Formats:** Use Parquet or ORC for efficient analytics.
- **Compression:** Reduce storage requirements and improve performance.
- **Caching:** Store frequently accessed data in memory.
- **File Optimization:** Maintain optimal file sizes for better processing.

Use Cases

- Big Data Analytics
- Machine Learning
- Real-Time Data Processing

- Data Science Projects

Business Benefits

- Real-time analytics
- Demand forecasting
- Customer behavior analysis
- Fraud detection

Challenges of Data Lakes

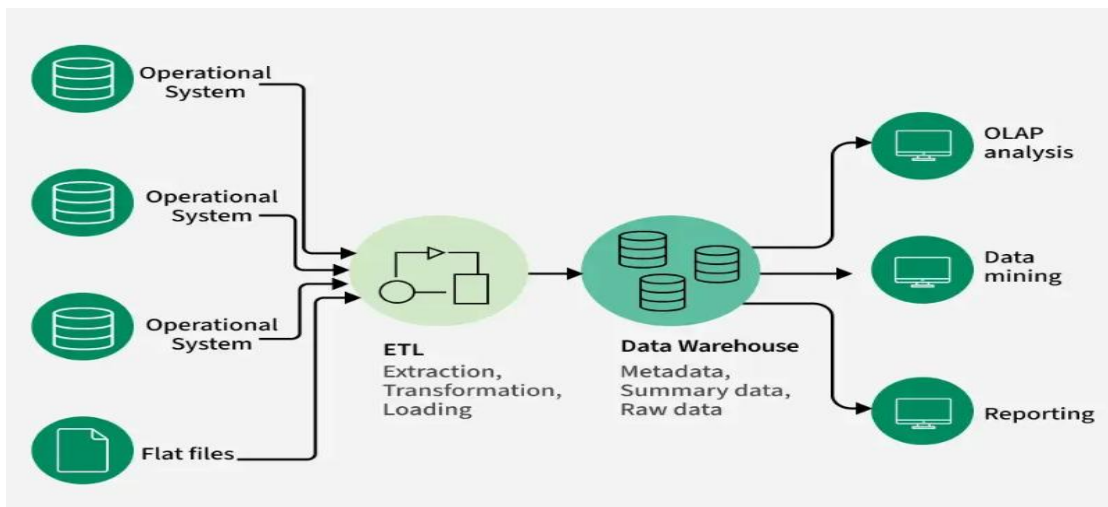
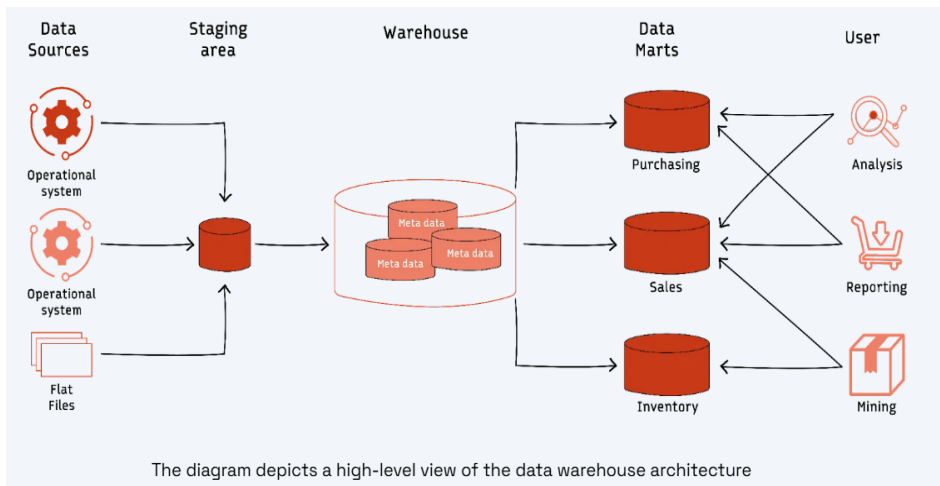
- Data Quality Issues
- Security and Access Control
- Metadata Management
- Integration Complexity
- Specialized Skill Requirements

Purpose

Store large volumes of raw data for future processing, analysis, and AI applications..

Data Warehousing

- Data warehousing is the process of collecting, integrating, storing and managing data from multiple sources in a central repository.
- It enables organisations to organise large volumes of current and historical data for efficient querying, analysis and reporting.
- A data warehouse helps businesses access data quickly for reporting and decision-making.



Need for Data Warehousing

a. Handling Large Data Volumes:

Stores and manages large amounts of data efficiently for analysis.

b. Enhanced Analytics

Optimized for reporting, analytics, and historical insights.

c. Centralized Data Storage

Combines data from multiple sources into a single view.

d. Trend Analysis

Uses historical data to identify trends and support forecasting.

e. Business Intelligence Support

Provides quick insights through BI tools for better decision-making.

Key Components of a Data Warehouse

Component	Description
Data Sources	Provide raw data from databases, applications, and external systems.
ETL Process	Extracts, cleans, transforms, and loads data into the warehouse.
Data Warehouse Database	Central repository that stores processed and organized data.
Metadata	Information that describes the structure and meaning of data.
Data Marts	Smaller data stores created for specific business teams or departments.
OLAP Tools	Enable multidimensional analysis and complex data exploration.
End-User Access Tools	Dashboards and BI tools used for reporting and decision-making.

Performance Optimization

- Query Optimization:** Improve data retrieval speed.
- Indexing:** Create shortcuts for faster searches.
- Data Partitioning:** Split large datasets into smaller sections.
- Data Sharding:** Distribute data across multiple servers for scalability.

Data Warehouse vs DBMS

Feature	Database	Data Warehouse
Purpose	Used for operational and transactional processing.	Used for analytical processing and decision-making.

Feature	Database	Data Warehouse
Data Type	Stores current and up-to-date data for daily operations.	Stores historical data collected over time.
Usage	Supports data entry, updates, and transactions.	Supports reporting, trend analysis, forecasting, and BI.
Scope	Application-specific and designed for a particular system.	Integrates data from multiple sources across the organization.
Data Structure	Optimized for fast insert, update, and delete operations.	Optimized for complex queries and data analysis.
Time Horizon	Focuses on present and recent data.	Maintains historical data for long-term analysis.
Example	School database storing student details, attendance, and marks.	Centralized repository analyzing school performance across a city.

Challenges in Building a Data Warehouse

Data Collection Strategy

- Data can be pushed from source systems or pulled by the warehouse on a schedule.
- Minor delays in synchronization are usually acceptable for analytics.

Schema Design and Integration

- Data from different sources must be standardized into a common format.
- A unified schema ensures consistency and easier analysis.

Data Cleaning and Transformation

- Data is cleaned to remove errors, inconsistencies, and duplicates.
- Transformation converts data into a format suitable for analysis.

Managing Data Updates

- Changes in source systems need to be reflected in the warehouse.
- Maintaining consistency during updates can be challenging.

Aggregation and Summarization

- Large datasets are often summarized to improve performance.
- Aggregated data enables faster reporting and analysis.

Real world Example of Data warehousing

Data Warehousing can be applied anywhere where we have a huge amount of data and we want to see statistical results that help in decision making.

1. E-commerce

- **Data Gathering:** Orders, returns, payments, user clicks, delivery updates.
- **Schema:** Combines source data into a structured star schema for analysis.
- **Cleansing:** Standardizes customer names, locations and product categories.
- **Updates:** Near real-time or scheduled loads for fresh insights.
- **Summarization:** Bestsellers by category, regional demand trends, logistics performance.

2. Banking

- **Data Gathering:** ATM transactions, online banking, credit card usage, loan records.
- **Schema:** Integrates data from core banking, CRM and fraud detection systems.
- **Cleansing:** Fixes inconsistencies in account info, transaction logs and addresses.
- **Updates:** Transaction data is batched and uploaded nightly.
- **Summarization:** Daily cash flow reports, high-risk account flags and customer profitability analysis.

Advantages

- **Better Decisions:** Centralized data supports faster, smarter decisions.
- **Business Intelligence:** Enables strong operational insights.

- **High Data Quality:** Ensures consistency and reliability.
- **Scalable:** Handles large and growing datasets.

Disadvantages

- **High Cost:** Setup requires major investment.
- **Complexity:** Needs skilled professionals to manage.
- **Time-Consuming:** Long setup and integration time.
- **Integration Issues:** Combining data from sources can be challenging.

Data Modelling

The process of organizing and structuring data to improve storage, retrieval, and analysis.

Purpose:

Create a logical framework that makes data easy to understand and use.

Data Pipelines

Automated workflows that move and process data from source systems to storage and analytics platforms.

Purpose:

Ensure data is delivered efficiently and reliably for business use.

Data Quality & Validation

The process of checking and improving data accuracy, completeness, and consistency.

Purpose:

Ensure trustworthy data for reporting, analytics, and AI applications.

Key Areas:

- **Data Accuracy:** Ensure data is correct and reliable.
- **Data Cleaning:** Remove errors, duplicates, and inconsistencies.
- **Data Validation:** Verify data meets predefined quality standards.
- **Data Governance:** Define policies for data ownership, security, and compliance.
- **Compliance & Security:** Protect data through access controls, encryption, and regulatory compliance.

Analytics & AI Applications

The use of prepared data to generate insights, build reports, and develop intelligent solutions.

Purpose:

Support data-driven decisions, predictions, and automation.

Data Engineering Tools and Technologies

Databases (SQL vs NoSQL)

SQL Databases

- Store structured data in tables with predefined schemas.
- Best suited for transactional applications and complex queries.
- **Examples:** MySQL, PostgreSQL, SQL Server.

NoSQL Databases

- Store semi-structured or unstructured data with flexible schemas.
- Ideal for large-scale and rapidly changing data.
- **Examples:** MongoDB, Cassandra, Redis.

Data Processing Frameworks

Tool	Purpose	Best Used For
Apache Hadoop	Distributed storage and processing of large datasets	Big Data processing and batch analytics
Apache Spark	Fast in-memory processing engine	Real-time analytics, ETL, and machine learning

Data Warehousing Solutions

Tool	Purpose	Best Used For
Amazon Redshift	Cloud-based data warehouse	Large-scale analytics on AWS
Google BigQuery	Serverless analytics platform	Fast querying of massive datasets
Snowflake	Cloud-native data warehouse	Scalable analytics across multiple clouds

Summary

- **Databases** → Store operational and application data.
- **Processing Frameworks** → Transform and process large volumes of data.
- **Data Warehouses** → Store analytics-ready data for reporting and business intelligence.
- Together, these tools form the foundation of modern **Data Engineering solutions**.

Workflow Orchestration Tools: Apache Airflow, Luigi

- **Apache Airflow**: An open-source platform for scheduling and monitoring workflows. It allows data engineers to define complex workflows with dependencies.
- **Luigi**: A Python package for building complex pipelines and managing dependencies between tasks.